Innovation, Demand, and Finance in an Agent Based-Stock Flow Consistent model

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Abstract

Keyword:

In this paper we present an Agent Based macroeconomic model with heterogeneous agents interacting through a decentralized matching process presenting common features across the markets. The economy is composed of:

- Households, selling workforce to firms, consuming and saving; their savings being allocated between cash and deposits.
- Firms, divided into Consumption and Capital firms. Consumption firms produce a consumption homogeneous good using labor and capital goods produced by capital firms. Capital firms produce different vintages of capital goods using labor. Capital firms invest in R&D and may thus come to produce more productive vintages of capital. Firms use both internal and external finance to fund their expenses for wages and investment (in capital accumulation and R&D for consumption and capital firms respectively. The amount of external finance, that is loans asked by firms, depends on the specific financial strategy they adopt. In particular, we consider two financial behaviors inspired by the "pecking order" and "dynamic trade-off" theories of finance. In the first case firms resort to external finance only when internal funding is not sufficient to cover the financial requirement. In the second case the shares of internal and external finance are chosen in order to attain a target leverage ratio, endogenously evolving over time.
- Banks, which collect deposits from households and grant loans to firms. In order to give a more realistic representation of banks' balance sheet active management, we also implement Basel III constraints regarding banks' capital, leverage and reserve ratios requirements.
- A Central Bank, which holds banks' reserve accounts and offer cash advances at the discount rate.
- A government sector, which hires public workers, collect taxes and issue bonds to cover the deficit between its current expenses (wages paid, interest on previously issued bonds and repayment of bonds reaching maturity) and taxes. Bonds are bought by banks and provide an interest which reflects the government financial reliability.

Agents locally interact on the good, labor, credit, deposit, and reserve markets through a common decentralized matching mechanism, following the logic explained in Riccetti et al. (2014). The investment by capital firms in R&D triggers a process of Schumpeterian competition in both the capital and consumption sectors as capital firms compete by producing capital goods which entails different levels of productivity, while consumption firms compete in an homogenous good market trying to lower their production costs through capital investment in newly created vintages. Therefore, firms may go bankrupt during the simulation, inducing a loss for banks related to the nonperforming loans, which in turn may result in bank's bankruptcy. The shock related to the endogenous default of firms may thus propagate in the economy through contagion effects across agents' balance sheets, possibly generating a cascade of failures which in turn affects employment and demand dynamics. Furthermore, by affecting labor productivity in the consumption sector, technological innovation may exert an impact on labor demand by consumption firms, thus possibly exerting an impact also on demand patterns. On the other hand, a weak demand can induce a drop in firms' sells, thus depress-

ing R&D investment and possibly affecting innovation dynamics. The model thus aims at analyzing the reciprocal influence of demand and supply factors in shaping economic cycles, building a bridge between the Keynesian and Schumpeterian school of thoughts. Finally, we believe the model implements for the very first time in the field of Innovation Economics Stock Flow Consistency (Godley and Lavoie, 2007) at both the micro and macroeconomic levels, by providing a detailed, comprehensive, and rigorous accounting of real and financial flows and stocks. This feature, as already explained in Caiani et al. (2014) is particularly relevant to analyze the inter-linkages between the real and financial sides of an economic system.

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